

THE DEGREE-SIX INTEGRAL CYCLE CLASS MAP OF $B(\mathrm{SL}_6/\mu_2)$ IS SURJECTIVE

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ABSTRACT. Let $G = \mathrm{SL}_6/\mu_2$ over $\mathbb{C}$. Tripathy established the chain of subgroups $2\mathbb{Z} = \mathrm{Im} \mathrm{CH}^3(BG) \subseteq \mathrm{Im} \mathrm{CH}^3(BG) \subseteq \mathrm{Im} MU^6(BG) = \mathbb{Z}$ inside $H^6(B\mathrm{SL}_6; \mathbb{Z}) = \mathbb{Z}c_3$ and asked whether $\mathrm{Im} \mathrm{CH}^3(BG)$ equals $\mathbb{Z}$ or $2\mathbb{Z}$; the latter case would be a counterexample, in the lowest open codimension, to the conjecture of Totaro (strengthened by Yagita) that $\mathrm{CH}^*(BG) \rightarrow MU^{2*}(BG) \otimes_{MU^*} \mathbb{Z}$ is an isomorphism. We resolve the question positively, by an explicit cycle: the Severi–Brauer transfer $x_w = \pi_*(c_4(V_1)^2) + 68c_3(\Lambda^2 \mathrm{std})$, where π is the Severi–Brauer scheme of the versal degree-six Azumaya algebra and V_1 its tautological rank-six bundle, satisfies $g^*(x_w) = c_3$ on the nose. Refining the computation two steps further, we show that the cycle class of x_w is exactly a free generator of $H^6(B(\mathrm{SL}_6/\mu_2); \mathbb{Z})$, and that the degree-six torsion class is the cycle class of an explicit algebraic cycle, the difference of a Chern class and twice the transfer witness; consequently the integral cycle class map $\mathrm{CH}^3(B(\mathrm{SL}_6/\mu_2)) \rightarrow H^6(B(\mathrm{SL}_6/\mu_2); \mathbb{Z})$ is surjective. In particular $\mathrm{CH}^3(BG)$ is not generated by Chern classes of representations, yet every integral cohomology class of degree six is algebraic.

1. INTRODUCTION

Among the quotients SL_n/μ_ℓ , the group $G = \mathrm{SL}_6/\mu_2$ occupies a distinguished position in the study of algebraic cycles on classifying spaces. Totaro [11] defined the Chow ring $\mathrm{CH}^*(BG)$ and the refined cycle class map

$$\mathrm{CH}^*(BG) \longrightarrow MU^{2*}(BG) \otimes_{MU^*} \mathbb{Z} \longrightarrow H^{2*}(BG; \mathbb{Z}),$$

and raised the possibility that the first map is an isomorphism for every complex algebraic group – immediately hedged: the map “is probably not always an isomorphism” [11]; Yagita [14, §12] strengthened the statement to the level of algebraic cobordism (see Remark 1.2 for the precise layering of these conjectures, which the literature often cites in a single clause). In codimension two the map $\mathrm{CH}^2(BG) \rightarrow MU^4(BG) \otimes_{MU^*} \mathbb{Z}$ is an isomorphism for every G [11, §3], so the known codimension-two failures of integral-Hodge-type statements on $B(\mathrm{SL}_n/\mu_\ell)$ [1, 13] – which occur at free classes – are already dictated by cobordism. Codimension three is the first place a free class can separate CH from MU , and Tripathy [13] isolated the

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critical instance. For $BG = B(\mathrm{SL}_6/\mu_2)$ he proved, with all images taken in $H^6(\mathrm{BSL}_6; \mathbb{Z}) = \mathbb{Z}c_3$ (his standing convention; see Convention 2.2):

$$2\mathbb{Z} = \mathrm{Im} Ch^3(BG) \subseteq \mathrm{Im} CH^3(BG) \subseteq \mathrm{Im} MU^6(BG) = \mathbb{Z},$$

where Ch^3 denotes the subgroup generated by Chern classes of representations. Thus topology (through complex cobordism, equivalently through connective K -theory) permits the full group $\mathbb{Z}$, representation theory supplies only $2\mathbb{Z}$, and the Chow group sits undetermined in between. Tripathy wrote that the question of whether $\mathrm{Im} CH^3(BG)$ is $\mathbb{Z}$ or $2\mathbb{Z}$ “is certainly of pressing interest; however, we cannot treat it with the tools of representation theory and algebraic topology we brought to bear in this paper, as we need some further legitimate geometry,” and proposed the orbit stratification of $\mathrm{Sym}^2(\mathbb{C}^6)$ as a candidate. In the decade since, the question has remained open: the projective-unitary literature at odd primes (motivic power operations, period–index arguments) is structurally silent on a two-primary free class, and the stratification programme, carried out by the author (unpublished), reproduces the even bound through its boundary contributions.

This paper resolves the question, positively, by geometry of a different twisted kind: the Severi–Brauer scheme of the tautological Azumaya algebra.

Theorem 1.1. *Let $G = \mathrm{SL}_6/\mu_2$ over $\mathbb{C}$, let X be a Totaro approximation of BG , let $\pi: P = \mathrm{SB}(\mathcal{A}) \rightarrow X$ be the Severi–Brauer scheme of the versal degree-six Azumaya algebra, $V_1 = \mathrm{Hom}(\mathcal{O}(-1), \mathcal{O} \otimes \mathbb{C}^6)$ its tautological rank-six bundle, and set*

$$x_w := \pi_*(c_4(V_1)^2) + 68c_3(\Lambda^2 \mathrm{std}) \in CH^3(BG).$$

Then:

- (1) $g^*(x_w) = c_3$, where $g: \mathrm{BSL}_6 \rightarrow BG$ is the covering map. Hence, in Tripathy’s normalization, $\mathrm{Im} CH^3(BG) = \mathbb{Z} \subset H^6(\mathrm{BSL}_6; \mathbb{Z})$: the dichotomy of [13, §5] resolves to the algebraic case, and no counterexample to the surjectivity half of the Totaro–Yagita conjectures arises from SL_6/μ_2 in codimension three (see Remark 5.3 for the precise scope).
- (2) $\mathrm{cl}(x_w)$ is a free generator of $H^6(BG; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/2$: the torsion component of $\mathrm{cl}(x_w)$ vanishes.
- (3) The degree-six torsion class $\tilde{x}_1^2$ (the square of the integral Brauer class; Lemma 2.4) equals $\mathrm{cl}(c_3(\Lambda^2 \mathrm{std}) - 2x_w)$. Consequently

$$\mathrm{cl}: CH^3(B(\mathrm{SL}_6/\mu_2)) \longrightarrow H^6(B(\mathrm{SL}_6/\mu_2); \mathbb{Z})$$

is surjective.

Part (1) is the resolution of Tripathy’s question. Parts (2) and (3) refine it to the strongest degree-six statement available: on these smooth quasi-projective approximations every integral cohomology class of degree six is pure Tate of Hodge type, so (3) says that *every* degree-six integral Hodge

class of the approximations is algebraic – the exact opposite of the counterexample the torsion examples of Atiyah–Hirzebruch type might have led one to expect. Since Chern classes of representations generate only $2\mathbb{Z}c_3$ on the free part, the witness x_w is certifiably exotic: $\mathrm{CH}^3(BG)$ is a Chow group of a classifying space that is provably not generated by Chern classes of representations (indeed not by Chern classes of any vector bundles on the approximations pulled from representations), yet nothing in H^6 escapes it.

The proof has three independent inputs, and we have kept them separated. The first (Section 3) is the construction and the unit certificate: the Severi–Brauer scheme of the versal algebra is itself a classifying-space approximation – of P_1/μ_2 , where P_1 is a parabolic of SL_6 – so its Chow ring is a ring of honest characteristic classes, and the transfer π_* reaches classes that representation theory cannot; a Segre-class computation on the split cover evaluates the transfer at $-135c_3$, an *odd* multiple, and one even Chern correction produces the exact unit. The second input (Section 4) is a vanishing: restricted along $B\mathrm{P}\mathrm{Sp}_6 \rightarrow BG$, the free generator dies while the torsion class survives, and the entire mod-two transfer of the restricted Severi–Brauer bundle vanishes by a two-line projection-formula argument resting on Quillen’s observation that a Severi–Brauer scheme splits its own algebra; this proves (2). The third input (Section 5) is a nonvanishing: restricted along $BSO(3) \rightarrow BG$ through the tensor embedding $\mathbb{C}^6 = \mathbb{C}^2 \otimes \mathbb{C}^3$, the mod-two class $c_3(\Lambda^2 \mathrm{std})$ is detected to be exactly the torsion class, which proves (3). The mod-two cohomological control throughout comes from a nineteen-seventy-five computation of Kono and Mimura [8] whose relevance to this circle of questions seems not to have been noticed: their presentation of $H^*(B(\mathrm{SU}(4n+2)/\{\pm I\}); \mathbb{F}_2)$ contains, at $n = 1$, both the relation ($a_3a_6 = 0$) that forces the normalization underlying (2) and the exterior-square representation underlying (3).

Remark 1.2 (Attribution of the conjectures). The conjecture tested here is usually cited, following [13, §§1, 5], as “Yagita’s Conjecture 12.2,” in the form: $\mathrm{CH}^*(BG) \rightarrow MU^{2*}(BG) \otimes_{MU^*} \mathbb{Z}$ is an isomorphism. The sources are more layered. Yagita’s §12 first records that Totaro conjectured this isomorphism (an unnumbered display in [14, p. 30]; the conjecture appears, hedged, in [11]), and then states as its numbered Conjecture 12.2 a strictly stronger, cobordism-level assertion: that the isomorphisms of his Theorem 12.1 – $MGL^{2*,*}(BG; \mathbb{Z}/p) \cong MU^{2*}(BG; \mathbb{Z}/p) \cong MU^{2*}(BG)/p$ together with τ -freeness of $MGL^{*,*'}(BG; \mathbb{Z}/p)$ – hold for every connected algebraic group. Totaro’s book confirms the layering: the Chow-level isomorphism is “as conjectured in [11],” which Yagita “strengthened ... to say that algebraic cobordism Ω^*BG should map isomorphically to the topologically defined MU^*BG ” [12, Preface]. The strengthened form implies the Chow-level one by Levine–Morel [9]. What the present paper proves is the degree-six cohomological consequence, at this instance, of the surjectivity half – the statement Tripathy’s dichotomy encodes – together with its

torsion-exact refinement. The cobordism-level Conjecture 12.2 at this instance (τ -freeness, and surjectivity onto the torsion of $MU^6(BG) \otimes_{MU^*} \mathbb{Z}$) is strictly stronger than its cohomological shadow and is *not* claimed here.

Remark 1.3 (Verification). The pushforward computations in this paper are short enough to check by hand, and we present them that way; they were additionally verified by computer algebra in exact arithmetic, in three independent implementations (direct Segre expansion, expansion over formal Chern roots, and torus-equivariant fixed-point localization at two generic integer points), together with the known-answer controls detailed in Remark 3.6.

2. NOTATION AND RECOLLECTIONS

Convention 2.1 (Classifying spaces and approximations). All schemes are of finite type over $\mathbb{C}$. For an affine algebraic group H , a *Totaro approximation* of BH (in codimension $\leq N$) is a quotient U/H , where U is an open H -stable subscheme of a representation W on which H acts freely and $\text{codim}_W(W \setminus U) > N$; then $\text{CH}^i(BH) := \text{CH}^i(U/H)$ for $i \leq N$ is independent of choices [11, 2], and likewise $H^i((U/H)^{\text{an}}; \mathbb{Z})$ computes $H^i(BH; \mathbb{Z})$ of the topological classifying space in the same range. We write $X = U/G$ for a fixed approximation of BG , $G = \text{SL}_6/\mu_2$, with $N > 16$, and we abbreviate cohomology of the analytification by $H^*(X; \mathbb{Z})$. All statements below are stable in N . The cycle class map $\text{cl}: \text{CH}^i(X) \rightarrow H^{2i}(X; \mathbb{Z})$ is the usual one [3, Ch. 19]; it commutes with flat pullback and with proper pushforward [3, §19.1].

Convention 2.2 (Tripathy’s normalization). Following [13, §2] (“whenever we consider the image of some group, we mean the image under the composition of the relevant cycle class map and pullback to the integral cohomology of BSL_n ”), for a cohomology theory E we write $\text{Im } E(BG)$ for the image of the composite $E^6(BG) \rightarrow H^6(BG; \mathbb{Z}) \xrightarrow{g^*} H^6(\text{BSL}_6; \mathbb{Z}) = \mathbb{Z}c_3$, where $g: \text{BSL}_6 \rightarrow BG$ is induced by the central isogeny. On approximations g is realized as follows: choose an auxiliary SL_6 -representation W with an open locus $W \setminus T$ on which SL_6 acts freely (large codimension); then $X' = (U \times (W \setminus T))/\text{SL}_6$ approximates BSL_6 and the projection $g: X' \rightarrow U/G = X$ is a fibration with fibre $(W \setminus T)/\mu_2$ – an approximation of $B\mu_2$ – hence a smooth, in particular flat, morphism.

We collect the cohomological inputs. Throughout, $H^*(\text{BSL}_6; \mathbb{Z}) = \mathbb{Z}[c_2, \dots, c_6]$ and $\text{CH}^*(\text{BSL}_6) = \mathbb{Z}[c_2, \dots, c_6]$ with cl matching c_i to c_i [11]; in particular $\text{CH}^3(\text{BSL}_6) = \mathbb{Z}c_3$ and $H^6(\text{BSL}_6; \mathbb{Z}) = \mathbb{Z}c_3$ is torsion-free.

Lemma 2.3 (Kono–Mimura, 1975). *Write $BG(6)$ for the classifying space of $\text{SU}(6)/\{\pm I\}$, the compact form of $B(\text{SL}_6/\mu_2)$. Then $H^*(BG(6); \mathbb{F}_2)$ admits the presentation of [8, Theorems 3.4 and 4.11]: it is generated by classes a_2, a_3 (degrees 2, 3), $a_6 = y(1)$, $a_{10} = y(2)$ (degrees 6, 10; the identification*

$y(\{i\}) = a_{4i+2}$ is [8, Remark 3.5]), x'_{16}, x'_{24} , and $y(1, 2)$ (degree 18), subject in degrees ≤ 15 only to the relations

$$a_3 a_6 = 0, \quad a_3 a_{10} = 0$$

of [8, Proposition 5.6(3)]. In particular:

- (1) $H^6(BG(6); \mathbb{F}_2)$ has basis $\{a_2^3, a_3^2, a_6\}$ and $H^7(BG(6); \mathbb{F}_2)$ has basis $\{a_2^2 a_3\}$;
- (2) the Steenrod actions $Sq^1 a_2 = a_3$ and $Sq^1 a_3 = 0$ hold ([8, Lemma 5.5] together with the identification of a_2, a_3 with the images of the universal classes under the degree- ≤ 4 isomorphism with $H^*(BSO(3); \mathbb{F}_2)$ established on [8, p. 711]);
- (3) a_2 is the class of the nonsplit μ_2 -gerbe $B\mathrm{SL}_6 \rightarrow BG$ (the unique nonzero element of $H^2(BG; \mathbb{F}_2) = \mathbb{F}_2$), and a_2, a_3 restrict to zero on $B\mathrm{SL}_6$.

The comparison of the algebraic $B(\mathrm{SL}_6/\mu_2)$ (approximations) with the compact $BG(6)$ is the standard one: $\mathrm{SU}(6) \subset \mathrm{SL}_6(\mathbb{C})$ is a maximal compact subgroup compatible with the central μ_2 , so the analytifications of the approximations map to $BG(6)$ by maps that are cohomology isomorphisms in the stable range. The same convention applies, without further comment, to every reductive pair used below ($\mathrm{Sp}_6/\mathrm{PSp}_6$ with $Sp(3)$, SO_3 with $SO(3)$, GSp_4 with its maximal compact), and a parabolic deformation-retracts onto its Levi, so classifying spaces of parabolic quotients may be computed on the Levi. We write the mod-two cohomology of X by the same names.

Proof. Everything is as cited; we make three glosses. First, the degree count: the generators in degrees ≤ 15 are a_2, a_3, a_6, a_{10} , and the ideal of relations is generated in degrees 9, 13, 16, 21, $\dots$ ($a_3 y(I)$, the $y(I)y(J)$ -relations, and the squares $y(I)^2$, of which the singleton instances are vacuous by the identification $y(\{i\}) = a_{4i+2}$); hence in degrees ≤ 8 the ring is free on the monomials in a_2, a_3, a_6 , giving (1). Second, $Sq^1 a_3 = Sq^1 Sq^1 a_2 = 0$. Third, (3): the fibration $B\mu_2 \rightarrow B\mathrm{SL}_6 \rightarrow BG$ is classified by a map $BG \rightarrow K(\mathbb{Z}/2, 2)$ that is an isomorphism on π_2 and, since $\pi_3(BG) = \pi_2(G) = 0$, induces an isomorphism on $H^i(-; \mathbb{F}_2)$ for $i \leq 3$; the class a_2 corresponds to the fundamental class ι_2 , whose transgression description is the statement that the gerbe is classified by a_2 , and a_2, a_3 pull back to $B\mathrm{SL}_6$ into $H^{2,3}(B\mathrm{SL}_6; \mathbb{F}_2) = 0$. $\square$

Lemma 2.4 (The degree-six integral cohomology of BG). $H^6(BG; \mathbb{Z}) \cong \mathbb{Z} \oplus \mathbb{Z}/2$. The torsion subgroup is generated by $\tilde{x}_1^2$, where $\tilde{x}_1 := \beta(a_2) \in H^3(BG; \mathbb{Z})$ is the integral Bockstein of the gerbe class ($2\tilde{x}_1 = 0$, $\mathrm{red}_2 \tilde{x}_1 = a_3$), and $\mathrm{red}_2(\tilde{x}_1^2) = a_3^2$. Moreover the mod-two reduction of any free generator of $H^6(BG; \mathbb{Z})$ lies in $\{a_6, a_6 + a_3^2\}$; there is consequently a unique-up-to-sign free generator $\tilde{e}$ with

$$\mathrm{red}_2(\tilde{e}) = a_6,$$

and we fix this normalization for the rest of the paper. Finally $g^*: H^6(BG; \mathbb{Z}) \rightarrow H^6(BSL_6; \mathbb{Z})$ kills the torsion and sends $\tilde{e}$ to a_{c_3} for some integer a (whose value ± 1 will follow from Theorem 1.1(1)).

Proof. Rationally, $H^*(BG; \mathbb{Q}) = H^*(BSL_6; \mathbb{Q})$ (the fibre $B\mu_2$ of $B\mu_2 \rightarrow BSL_6 \rightarrow BG$ is $\mathbb{Q}$ -acyclic, so the Serre spectral sequence of the fibration collapses and g^* is an isomorphism with $\mathbb{Q}$ -coefficients – and likewise with $\mathbb{Z}[1/2]$ -coefficients), so $H^6(BG; \mathbb{Q}) = \mathbb{Q}$ and all torsion is 2-primary. For the 2-primary torsion we use the Bockstein spectral sequence of the ring of Lemma 2.3. In degree six, $\ker Sq^1 = \langle a_3^2, a_6 \rangle$ (as $Sq^1(a_3^2) = a_2^2 a_3 \neq 0$, $Sq^1(a_3^2) = 0$, and $Sq^1 a_6 = 0$ – the last because $0 = Sq^1(a_3 a_6) = a_3 Sq^1 a_6$ while $a_3 \cdot a_2^2 a_3 = a_2^2 a_3^2 \neq 0$ excludes the only degree-seven candidate) and $\text{Im}(Sq^1: H^5 \rightarrow H^6) = \langle a_3^2 \rangle$ (from $Sq^1(a_2 a_3) = a_3^2$; H^5 has basis $\{a_2 a_3\}$). So the first Bockstein page in degree six is $\langle a_6 \rangle$, of dimension one, matching the rational rank; hence the torsion of $H^6(BG; \mathbb{Z})$ has exponent two and its rank equals $\dim \text{Im}(Sq^1: H^5 \rightarrow H^6) = 1$. The class $\tilde{x}_1^2 = \beta(a_2)^2$ is a nonzero order-two class with reduction a_3^2 (as $\text{red}_2 \beta = Sq^1$ and reduction is a ring map), so it generates. For the normalization: the reduction of an integral class is a Sq^1 -cycle, which by the displayed kernel excludes any a_2^3 -component; and the a_6 -coefficient of the reduction of a free generator must be one – otherwise the reduction lies in $\langle a_3^2 \rangle = \text{red}_2(\text{tors})$, so the generator is congruent modulo $2H^6$ to a torsion class and is therefore divisible by two in $H^6/\text{tors} \cong \mathbb{Z}$, contradicting generation. So the reduction is a_6 or $a_6 + a_3^2$, and replacing $\tilde{e}$ by $\tilde{e} + \tilde{x}_1^2$ toggles between them. The last sentence is immediate since $H^6(BSL_6; \mathbb{Z})$ is torsion-free. $\square$

Lemma 2.5 (The adjoint symplectic group). *Let $\text{PSp}_6 = \text{Sp}_6/\mu_2$ (rank three, adjoint of type C_3), embedded in G via $\text{Sp}_6 \subset \text{SL}_6$ (centres matched). Then:*

- (1) $H^*(B\text{PSp}_6; \mathbb{F}_2) = \mathbb{F}_2[t_2, t_3, t_8, t_{12}]$, a polynomial ring on generators of the indicated degrees [4];
- (2) $Sq^1 t_2 = t_3$; the restriction $H^*(BG; \mathbb{F}_2) \rightarrow H^*(B\text{PSp}_6; \mathbb{F}_2)$ sends $a_2 \mapsto t_2$ and $a_3 \mapsto t_3$;
- (3) $H^6(B\text{PSp}_6; \mathbb{Z}) \cong \mathbb{Z}/2$, generated by W_3^2 where $W_3 = \beta(t_2)$, and reduction mod two is injective on it, with $\text{red}_2(W_3^2) = t_3^2$;
- (4) $\tilde{e}|_{B\text{PSp}_6} = 0$ and $\tilde{x}_1^2|_{B\text{PSp}_6} = W_3^2 \neq 0$.

Proof. (1) is Kono's determination [5], as recorded (with these generator names) in [4, §2.6, App. A]; the Steenrod action used in (2) is due to Stolz, as presented there. For (2): as in Lemma 2.3(3), $B\text{PSp}_6 \rightarrow K(\mathbb{Z}/2, 2)$ is an isomorphism on $H^{\leq 3}(-; \mathbb{F}_2)$ (again $\pi_2 = \mathbb{Z}/2$, $\pi_3 = 0$), so $t_3 = Sq^1 t_2$ is forced, being the image of $Sq^1 \iota_2 \neq 0$; and the extension $\mu_2 \rightarrow \text{SL}_6 \rightarrow G$ restricts along $\text{Sp}_6 \subset \text{SL}_6$ to the nonsplit extension $\mu_2 \rightarrow \text{Sp}_6 \rightarrow \text{PSp}_6$ (Sp_6 is a connected double cover), so the gerbe class a_2 restricts to the gerbe class t_2 , and $a_3 = Sq^1 a_2 \mapsto Sq^1 t_2 = t_3$.

(3): rationally $H^*(B\mathrm{PSp}_6; \mathbb{Q}) = \mathbb{Q}[p_1, p_2, p_3]$ with generators in degrees 4, 8, 12, so H^6 is all torsion, 2-primary by the same $\mathbb{Z}[1/2]$ -acyclicity argument as before. Bockstein counting in the polynomial ring: in degree six, $\ker Sq^1 = \langle t_3^2 \rangle$ ($Sq^1 t_2^3 = t_2^2 t_3 \neq 0$) and $\mathrm{Im}(Sq^1: H^5 \rightarrow H^6) = \langle t_3^2 \rangle$ ($Sq^1(t_2 t_3) = t_3^2$), so the first Bockstein page vanishes in degree six; the torsion has exponent two and rank $\dim \mathrm{Im}(Sq^1: H^5 \rightarrow H^6) = 1$. The class W_3^2 has order two and reduction $(Sq^1 t_2)^2 = t_3^2 \neq 0$, so it generates, and injectivity of red_2 on $H^6(B\mathrm{PSp}_6; \mathbb{Z}) \cong \mathbb{Z}/2$ is the nonvanishing of that reduction.

(4): $\tilde{x}_1 = \beta(a_2) \mapsto \beta(t_2) = W_3$ by (2) and naturality of the Bockstein, giving the second statement. For the first: $\mathrm{red}_2(\tilde{e}) = a_6$ restricts by (2) and Lemma 2.3 to a class killed by t_3 ($t_3 \cdot a_6| = (a_3 a_6)| = 0$), and t_3 is a non-zero-divisor in the polynomial ring, so $a_6|_{B\mathrm{PSp}_6} = 0$; hence $\tilde{e}|_{B\mathrm{PSp}_6}$ is an integral class with vanishing mod-two reduction, and by (3) it vanishes. $\square$

3. THE SEVERI–BRAUER TRANSFER AND THE UNIT CERTIFICATE

The central extension $\mu_2 \rightarrow \mathrm{SL}_6 \rightarrow G$ makes $\mathbb{C}^6$ a representation only of the double cover; descending $\mathrm{End}(\mathbb{C}^6)$, on which the centre acts trivially, yields a sheaf of Azumaya algebras $\mathcal{A}$ of degree six on X – the versal algebra – and its Severi–Brauer scheme $\pi: P = \mathrm{SB}(\mathcal{A}) \rightarrow X$ is a smooth proper étale-locally trivial $\mathbb{P}^5$ -bundle.

Proposition 3.1. *Let $P_1 \subset \mathrm{SL}_6$ be the parabolic stabilizer of a line in $\mathbb{C}^6$; note $\mu_2 \subset P_1$. Then $P = U/(P_1/\mu_2)$, the fibration induced by $P_1/\mu_2 \subset G$ with fibre $G/(P_1/\mu_2) = \mathrm{SL}_6/P_1 = \mathbb{P}^5$; in particular P is itself a Totaro approximation of $B(P_1/\mu_2)$ – a genuine classifying space, so every representation of P_1/μ_2 contributes an honest vector bundle on P with honest Chern classes. Moreover:*

- (1) *The assignment $V_1 := \mathrm{Hom}(\mathcal{O}(-1), \mathcal{O} \otimes \mathbb{C}^6)$ – the descent of the P_1 -representation $\mathrm{Hom}(\ell, \mathbb{C}^6)$, on which the centre acts by $(\pm 1)^{-1}(\pm 1) = +1$ – is an honest rank-six vector bundle on P , with a canonical nowhere-vanishing section ($c_6(V_1) = 0$) and fibre restriction $V_1|_{\mathbb{P}^5} = \mathcal{O}(1)^{\oplus 6}$.*
- (2) *If $g: X' \rightarrow X$ denotes the covering approximation of $B\mathrm{SL}_6$ and E' the (honest) descent of $\mathbb{C}^6$ to X' , then the square*

$$\begin{array}{ccc} \mathbb{P}(E') & \xrightarrow{g'} & P \\ \pi' \downarrow & & \downarrow \pi \\ X' & \xrightarrow{g} & X \end{array}$$

is Cartesian, $g'^ V_1 = \mathcal{O}(1) \otimes E'$, and $g^* \pi_* = \pi'_* g'^*$ on Chow groups.*

Proof. Étale-locally $\mathcal{A} \cong \mathrm{End}(\mathbb{C}^6)$ and the fibre of $\mathrm{SB}(\mathcal{A})$ is $\mathbb{P}(\mathbb{C}^6)$, on which G acts transitively through its quotient action with line-stabilizer P_1/μ_2 ; the mixing identification $U \times^G (G/(P_1/\mu_2)) = U/(P_1/\mu_2)$ gives the displayed description, and freeness plus the codimension bound make it an approximation of $B(P_1/\mu_2)$ [11, 2]. (1) is the displayed central-character computation

together with the tautological inclusion $\ell \hookrightarrow \mathbb{C}^6$, which is P_1 -equivariant and nowhere zero; along the fibre, $\mathrm{Hom}(\mathcal{O}(-1), \mathbb{C}^6) = \mathcal{O}(1)^{\oplus 6}$. For (2): over X' the algebra splits, $\mathcal{A}|_{X'} = \mathrm{End}(E')$, and Severi–Brauer formation commutes with base change, identifying the pullback of P with $\mathbb{P}(E')$ compatibly with the tautological data, whence $g'^*V_1 = \mathrm{Hom}(\mathcal{O}(-1), E') = \mathcal{O}(1) \otimes E'$. The base-change formula is [3, Prop. 1.7], applicable since g is smooth, hence flat (Convention 2.2), and π is proper. $\square$

Everything is now computable on the split side. Recall the Segre convention $c(E')s(E') = 1$, so that $\pi'_*(\xi^{5+i}) = s_i(E')$ for $\xi = c_1(\mathcal{O}(1))$ on $\mathbb{P}(E')$; with $c_1(E') = 0$ one has $s_1 = 0$, $s_2 = -c_2$, $s_3 = -c_3$.

Proposition 3.2. $\pi'_*(c_4(\mathcal{O}(1) \otimes E')^2) = 225 s_3(E') + 90 c_3(E') = -135 c_3$. Hence $y := \pi_*(c_4(V_1)^2) \in \mathrm{CH}^3(BG)$ satisfies $g^*(y) = -135 c_3$.

Proof. With $c_1(E') = 0$,

$$c_4(\mathcal{O}(1) \otimes E') = \binom{6}{4} \xi^4 + \binom{4}{2} \xi^2 c_2 + \binom{3}{1} \xi c_3 + c_4 = 15 \xi^4 + 6 \xi^2 c_2 + 3 \xi c_3 + c_4.$$

In the square, the only monomials whose pushforward lands in degree three are $225 \xi^8 \mapsto 225 s_3$ and $2 \cdot 15 \cdot 3 \xi^5 c_3 \mapsto 90 c_3$; the terms $180 \xi^6 c_2 \mapsto 180 s_1 c_2 = 0$ and all remaining cross terms push to Segre classes of negative index. Now $s_3 = -c_3$. $\square$

Proposition 3.3. *The exterior square $\Lambda^2 \mathrm{std}$ is an honest G -representation of rank fifteen, and on the cover $c_3(\Lambda^2 E') = 2 c_3$. Indeed, for six Chern roots $t_1, \dots, t_6$ one has the exact expansion*

$$e_3(\{t_i + t_j\}_{i < j}) = 10 e_1^3 + 16 e_1 e_2 + 2 e_3,$$

and $e_1 = c_1(E') = 0$. (For a bundle of general rank n with $c_1 = 0$ the analogous coefficient of e_3 is $n - 4$; only the value 2 at $n = 6$ is used in this paper, until Remark 5.4.)

Proof. The centre acts on $\Lambda^2 \mathbb{C}^6$ by $(\pm 1)^2 = +1$, so the representation descends to G . The displayed identity is a finite symmetric-function computation at $n = 6$ (verified symbolically, with zero residual at independent evaluation points; the rank-dependence of the coefficients is real – at general rank the e_3 -coefficient is $n - 4$ – which is why we state the identity at the rank where it is used). $\square$

Theorem 3.4 (The unit certificate). *Set $x_w := \pi_*(c_4(V_1)^2) + 68 c_3(\Lambda^2 \mathrm{std}) \in \mathrm{CH}^3(BG)$. Then*

$$g^*(x_w) = (-135 + 68 \cdot 2) c_3 = c_3.$$

Consequently:

- (1) *In Tripathy’s normalization (Convention 2.2), $\mathrm{Im} \mathrm{CH}^3(BG) \ni c_3$, so $\mathrm{Im} \mathrm{CH}^3(BG) = \mathbb{Z} c_3$: Theorem 1.1(1) holds.*

- (2) Writing $\mathrm{cl}(x_w) = \mu \tilde{e} + (\text{torsion})$ in $H^6(BG; \mathbb{Z})$ and $g^* \tilde{e} = a c_3$ as in Lemma 2.4, one has $\mu a = 1$ in $\mathbb{Z}$; hence $\mu = a = \sigma \in \{\pm 1\}$: the free component of $\mathrm{cl}(x_w)$ is a generator, and the restriction index of the free generator is one.

Proof. The formula assembles Propositions 3.2 and 3.3. (1) is immediate. For (2), apply g^* to $\mathrm{cl}(x_w) = \mu \tilde{e} + t$: the torsion t dies in the torsion-free $H^6(B\mathrm{SL}_6; \mathbb{Z})$, and $\mathrm{cl}(g^* x_w) = \mathrm{cl}(c_3) = c_3$ by naturality of cl under the flat g^* and the identification of cl on $B\mathrm{SL}_6$; so $c_3 = \mu a c_3$. (We emphasize that only flat-pullback naturality of cl is used here.) $\square$

Remark 3.5. By [13, §5], Chern classes of representations of G generate exactly $2\mathbb{Z}c_3 \subset H^6(B\mathrm{SL}_6; \mathbb{Z})$; the value $g^*(y) = -135c_3$ is odd, so the transfer class y – and a fortiori x_w – is not a polynomial in Chern classes of representations. The transfer evades the parity constraint because it does not factor through the representation ring: it is the pushforward of a characteristic class of the group P_1/μ_2 , which is not the restriction of a virtual representation of G .

Remark 3.6 (Numerical checks). Three controls, each capable of falsifying the computation, were run. (i) The codimension-two analogue: pushing forward all fourteen degree-7 monomials in the $c_i(V_1)$ produces values lying in $4\mathbb{Z}c_2$ without exception (greatest common divisor 24 over this set), consistent with the known $H^4(B(\mathrm{SL}_6/\mu_2); \mathbb{Z}) = \mathbb{Z} \cdot 4c_2$, Chern-generated [1, Prop. 2.2] – the transfer detects nothing new in codimension two, as it must not. (ii) The control group: for $\mathrm{SL}_4/\mu_2 = \mathrm{SO}_6$, where the codimension-three question is settled with index four, every transfer value is even – consistently with the general fact that the e_3 -coefficient of $c_3(\Lambda^2)$ vanishes at rank four. (iii) $\pi_*(c_5(V_1)) = \chi(\mathbb{P}^5) = 6$, and $c_6(V_1) = 0$ pushes to zero. (iv) Convention robustness: the four combinations of the two sign conventions (tautological line versus its dual; Segre classes of E' versus $E'^\vee$) produce the value set $\{-135, -315, +315, +135\}$ in Proposition 3.2, and the fixed-point localization implementation – which computes the pushforward without expanding in either convention – pins -135 .

4. THE TORSION COMPONENT OF THE WITNESS VANISHES

We now prove Theorem 1.1(2): in the normalization of Lemma 2.4, $\mathrm{cl}(x_w) = \sigma \tilde{e}$ exactly. Since $\mathrm{tors} H^6(BG; \mathbb{Z}) = \mathbb{Z}/2 \cdot \tilde{x}_1^2$ and 68 is even (so the Chern correction contributes no torsion: $68 \mathrm{cl}(c_3(\Lambda^2 \mathrm{std}))$ has torsion component $68 \cdot (\text{an order-two class}) = 0$), it suffices to show

$$\mathrm{cl}(y)|_{B\mathrm{PSp}_6} = 0 \quad \text{in} \quad H^6(B\mathrm{PSp}_6; \mathbb{Z}) = \mathbb{Z}/2 \cdot W_3^2,$$

because by Lemma 2.5(4) the restriction reads off exactly the torsion bit: $\mathrm{cl}(y) = \mu' \cdot 0 + \varepsilon W_3^2$ where $\varepsilon \in \{0, 1\}$ is the $\tilde{x}_1^2$ -component of $\mathrm{cl}(y)$.

Proposition 4.1. *Let $\pi^{\text{Sp}}: P^{\text{Sp}} \rightarrow X^{\text{Sp}}$ denote the restriction of the Severi–Brauer bundle along (an approximation of) $B\text{PSp}_6 \rightarrow BG$. Then the mod-two Gysin homomorphism $\pi_*^{\text{Sp}}: H^*(P^{\text{Sp}}; \mathbb{F}_2) \rightarrow H^{*-10}(B\text{PSp}_6; \mathbb{F}_2)$ vanishes identically. Consequently $\text{cl}(y)|_{B\text{PSp}_6} = \pi_*^{\text{Sp}}(\text{cl}(c_4(V_1^{\text{Sp}})^2)) = 0$ and $\varepsilon = 0$:*

$$\text{cl}(x_w) = \sigma \tilde{e} \quad \text{exactly in } H^6(BG; \mathbb{Z}).$$

Proof. First the identifications. The map of approximations $\phi: X^{\text{Sp}} = U/\text{PSp}_6 \rightarrow U/G = X$ (same U ; the G -action restricted to PSp_6 is still free) is smooth with fibre G/PSp_6 , hence flat, and Severi–Brauer formation commutes with base change, so $P^{\text{Sp}} = P \times_X X^{\text{Sp}}$ and, by [3, Prop. 1.7] again, $y|_{B\text{PSp}_6} = \pi_*^{\text{Sp}}(c_4(V_1^{\text{Sp}})^2)$ with V_1^{Sp} the pulled-back tautological bundle. Applying cl , which commutes with flat pullback and with proper pushforward [3, §19.1], reduces everything to the topological Gysin map of the $\mathbb{P}^5$ -fibration π^{Sp} . (Alternatively: the entire argument below may be run in topology from the start, since the class $\text{cl}(y)$ is by construction a topological transfer; the Chow-level statements ride along.)

Now the vanishing, in two lines. The pullback of the algebra to its own Severi–Brauer scheme splits: $\mathcal{A}|_{\text{SB}(\mathcal{A})} \cong \text{End}(J)^{\text{op}}$ (equivalently $\text{End}(J^V)$) for Quillen’s tautological bundle J – the display preceding Theorem 4.1 in [10, §8.4], classically Châtelet’s theorem over a field. Hence the Brauer class of $\mathcal{A}$ dies on P^{Sp} . The cohomological Brauer obstruction of an Azumaya algebra is the Bockstein of its gerbe class (Grothendieck [6]); over $B\text{PSp}_6$ the gerbe class of $\mathcal{A}$ is t_2 by Lemma 2.5(2), so in mod-two terms the class $t_3 = \text{red}_2 W_3 = \text{red}_2 \beta(t_2)$ pulls back to red_2 of the Brauer obstruction of a split algebra, i.e.

$$\pi^{\text{Sp}*}(t_3) = 0.$$

(Equivalently and independently: P^{Sp} is again homogeneous – Sp_6 acts transitively on $\mathbb{P}^5$, every line being isotropic for an alternating form, with line-stabilizer a parabolic whose Levi is $U(1) \times \text{Sp}_4 \ni -I$ – so $P^{\text{Sp}} \simeq B((U(1) \times \text{Sp}_4)/\Delta\mu_2) \cong B\text{GSp}_4$ via $(t, A) \mapsto tA$, and $H^3(B\text{GSp}_4; \mathbb{F}_2) = 0$ because the Serre spectral sequence of $B\text{Sp}_4 \rightarrow B\text{GSp}_4 \rightarrow BU(1)$, induced by $(t, A) \mapsto t^2$ with kernel Sp_4 , has base and fibre concentrated in even degrees.) By the projection formula, for every $x \in H^*(P^{\text{Sp}}; \mathbb{F}_2)$,

$$t_3 \cdot \pi_*^{\text{Sp}}(x) = \pi_*^{\text{Sp}}(\pi^{\text{Sp}*}(t_3) \cdot x) = 0,$$

and t_3 is a non-zero-divisor in the polynomial ring $H^*(B\text{PSp}_6; \mathbb{F}_2) = \mathbb{F}_2[t_2, t_3, t_8, t_{12}]$ (Lemma 2.5(1)); therefore $\pi_*^{\text{Sp}}(x) = 0$ for all x .

Finally, reduction mod two commutes with the Gysin map (both are induced, coefficient-naturally, by Poincaré duality and pushforward in Borel–Moore homology on the smooth quasi-projective approximations), and red_2 is injective on $H^6(B\text{PSp}_6; \mathbb{Z}) = \mathbb{Z}/2W_3^2$ (Lemma 2.5(3)); so the integral class $\text{cl}(y)|_{B\text{PSp}_6}$ vanishes, $\varepsilon = 0$, and by the reduction in the paragraph preceding the proposition, $\text{cl}(x_w) = \sigma \tilde{e}$ integrally – the torsion of $H^6(BG; \mathbb{Z})$ being 2-primary (Lemma 2.4), no odd-torsion component can hide. $\square$

Remark 4.2. The vanishing is wholesale: *no* class transferred from the Severi–Brauer bundle carries mod-two information over $B\mathrm{P}\mathrm{Sp}_6$ – the $\mathbb{P}^5$ -bundle analogue of the classical nonsplit period–index mechanism. It is instructive to see why the identical argument does not annihilate the transfer upstairs and so contradict Proposition 3.2: on BG the reduction a_3 of the Brauer class is a *zero-divisor* – the Kono–Mimura relation $a_3a_6 = 0$ – so the projection formula constrains π_* only into the annihilator of a_3 , which is exactly where $a_6 = \mathrm{red}_2(\tilde{e})$ lives; over $B\mathrm{P}\mathrm{Sp}_6$ the corresponding class t_3 is regular and the constraint is fatal. One relation from 1975 governs both halves of the computation.

5. THE TORSION CLASS IS ALGEBRAIC

We now prove Theorem 1.1(3). By Proposition 3.3 and Lemma 2.4 we may write $\mathrm{cl}(c_3(\Lambda^2\mathrm{std})) = 2u\tilde{e} + \delta\tilde{x}_1^2$ with $u \in \mathbb{Z}$ and $\delta \in \{0, 1\}$; applying g^* and Theorem 3.4(2) gives $2c_3 = 2u\sigma c_3$, so $u = \sigma$. The bit δ , unlike the ε of the previous section, is independent of the choice of free generator, since a change $\tilde{e} \mapsto \tilde{e} + \tilde{x}_1^2$ perturbs the formula by $2u\tilde{x}_1^2 = 0$.

Proposition 5.1. $\mathrm{red}_2(c_3(\Lambda^2\mathrm{std})) = a_3^2$ in $H^6(BG; \mathbb{F}_2)$; that is, $\delta = 1$.

Proof. The reduction lies in the span of $\{a_2^3, a_3^2, a_6\}$ (Lemma 2.3); we first pin its shape, then detect the coefficient. The a_6 -coefficient vanishes: restricting to $B\mathrm{SL}_6$, the left side becomes $\mathrm{red}_2(2c_3) = 0$, while $a_6 = \mathrm{red}_2(\tilde{e})$ restricts to $\mathrm{red}_2(\sigma c_3) = \bar{c}_3 \neq 0$ by Theorem 3.4(2), and a_2^3, a_3^2 restrict to zero; this forces the a_6 -coefficient to be zero. The a_2^3 -coefficient vanishes because the reduction of an integral class is a Sq^1 -cycle and Sq^1 kills a_3^2, a_6 but not a_2^3 (Lemma 2.4’s computation). So $\mathrm{red}_2 c_3(\Lambda^2\mathrm{std}) = \delta a_3^2$, and it remains to detect δ on a subgroup where a_3^2 survives.

Embed $\mathrm{SU}(2) = \mathrm{Sp}_2 \rightarrow \mathrm{SL}_6$ by $q \mapsto q \otimes I_3$ on $\mathbb{C}^6 = \mathbb{C}^2 \otimes \mathbb{C}^3$ (a symplectic form tensor a symmetric form is alternating); then $-I_2 \mapsto -I_6$, so the map descends to $\mathrm{SO}(3) = \mathrm{SU}(2)/\{\pm 1\} \rightarrow G$, and the pulled-back extension is the nonsplit $\mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$. Consequently, exactly as in Lemma 2.5(2), the gerbe class restricts as $a_2 \mapsto w_2$, hence $a_3 \mapsto Sq^1 w_2 = w_3$, in $H^*(BSO(3); \mathbb{F}_2) = \mathbb{F}_2[w_2, w_3]$; and a_6 restricts to zero, by the same two-step as before ($t \mapsto w$): $w_3 \cdot a_6| = (a_3a_6)| = 0$ and w_3 is a non-zero-divisor. The representation decomposes as

$$\Lambda^2(\mathbb{C}^2 \otimes \mathbb{C}^3) = (\Lambda^2\mathbb{C}^2 \otimes S^2\mathbb{C}^3) \oplus (S^2\mathbb{C}^2 \otimes \Lambda^2\mathbb{C}^3) = 6 \cdot \mathbf{1} \oplus 3 \cdot V_3,$$

where $V_3 = S^2\mathbb{C}^2$ is the complexification of the standard real representation R_3 of $\mathrm{SO}(3)$. For any real bundle R , the complexification satisfies $\mathrm{red}_2 c_l(R \otimes \mathbb{C}) = w_l(R)^2$: indeed $(R \otimes \mathbb{C})_{\mathbb{R}} \cong R \oplus R$, so the total Stiefel–Whitney class of the underlying real bundle is $w(R)^2 = \sum_l w_l(R)^2$ (squares have no cross terms mod two), and $\mathrm{red}_2 c_l = w_{2l}$ of the realification. Hence

$$\mathrm{red}_2 c(\Lambda^2\mathrm{std})|_{BSO(3)} = (1 + w_2^2 + w_3^2)^3,$$

whose degree-six component is $3w_3^2 = w_3^2 \neq 0$. Matching against $(\delta a_3^2)| = \delta w_3^2$ gives $\delta = 1$ (and, as a consistency check, the absence of any w_2^3 -term on the left matches the vanishing of the a_2^3 -coefficient established above). $\square$

Proof of Theorem 1.1. Part (1) and the unit normalization are Theorem 3.4. Part (2) is Proposition 4.1. For (3): by Propositions 4.1 and 5.1 and $u = \sigma$ (Section 5),

$$\text{cl}(c_3(\Lambda^2 \text{std}) - 2x_w) = (2\sigma \tilde{e} + \tilde{x}_1^2) - 2\sigma \tilde{e} = \tilde{x}_1^2,$$

so the image of cl contains $\tilde{e}$ and $\tilde{x}_1^2$. By Lemma 2.4 these generate $H^6(BG; \mathbb{Z}) \cong \mathbb{Z}\tilde{e} \oplus \mathbb{Z}/2\tilde{x}_1^2$, whence surjectivity – integrally, no localization required. $\square$

Corollary 5.2. *The Severi–Brauer transfer is defined natively on $BPU_6 = \text{BPGL}_6$: the versal algebra $\mathcal{A}$ and the tautological bundle V_1 descend (both have trivial central character for the full centre μ_6), so $y := \pi_*(c_4(V_1)^2) \in \text{CH}^3(BPU_6)$. Writing $H^6(BPU_6; \mathbb{Z}) = \mathbb{Z}e_3 \oplus \mathbb{Z}/2x_1^2$ [7, Thm. 1.1], with e_3 normalized as in Lemma 2.4 (Kono–Mimura’s Theorem 4.12 gives $BPU(6)$ the same mod-two presentation as $BG(6)$, so the normalization and its properties carry over verbatim), one has*

$$\text{cl}(y) = \mu' e_3 \quad \text{with } \mu' \text{ odd.}$$

In particular the free part of $H^6(BPU_6; \mathbb{Z})$ consists of algebraic classes after localization at two.

Proof. The centre $\mu_6 \subset \text{SL}_6$ acts on ℓ and on $\mathbb{C}^6$ by the same character, so it acts trivially on $\text{End}(\mathbb{C}^6)$ and on $\text{Hom}(\ell, \mathbb{C}^6)$: the constructions of Proposition 3.1 descend along $\text{SL}_6 \rightarrow \text{PGL}_6$, and the same base-change computation gives $g_{\text{PU}}^*(y) = -135c_3$ for the covering $g_{\text{PU}}: \text{BSL}_6 \rightarrow BPU_6$. Writing $e_3|_{\text{BSL}_6} = \lambda c_3$ ($\lambda \in \mathbb{Z}$; the target is torsion-free) and $\text{cl}(y) = \mu' e_3 + \varepsilon x_1^2$, restriction gives $\mu' \lambda = -135$; every divisor of 135 is odd, so μ' is odd. For ε : the argument of Proposition 4.1 runs verbatim over BPU_6 – the restriction of the μ_6 -gerbe class along $\text{PSP}_6 \subset \text{PU}_6$ is t_2 (the preimage of PSP_6 in SL_6 is $\mu_6 \cdot \text{Sp}_6$, and the two Bockstein ladders are compatible under $\mu_2 \hookrightarrow \mu_6$, the middle multiplication by three not affecting the connecting map), the relation $a_3 a_6 = 0$ holds in the $BPU(6)$ -presentation, and the restricted Severi–Brauer bundle is the same P^{Sp} – so $\text{cl}(y)|_{B\text{PSP}_6} = 0$ forces $\varepsilon = 0$. Since μ' is a 2-local unit, $e_3 \in \text{Im}(\text{cl}) \otimes \mathbb{Z}_{(2)}$.

We do not assert the BPU_6 -analogue of Theorem 1.1(3): the cycle $c_3(\Lambda^2 \text{std})$ does *not* descend to BPU_6 (the centre acts on $\Lambda^2 \mathbb{C}^6$ through μ_3), and transporting the torsion statement across the μ_3 -gerbe $B(\text{SL}_6/\mu_2) \rightarrow BPU_6$ – a 2-local equivalence on cohomology – would require a two-local comparison of Chow groups under the isogeny, which we neither need nor prove here. $\square$

Remark 5.3 (What is not claimed). Theorem 1.1 proves the degree-six *cohomological* consequence of the surjectivity half of the Totaro–Yagita conjectures at this instance, in its strongest (torsion-exact, integrally surjective) form. It does not prove the cobordism-level Conjecture 12.2 of [14] at this

instance: τ -freeness of $MGL^{*,*'}(BG; \mathbb{Z}/2)$, and surjectivity of $\mathrm{CH}^3(BG) \rightarrow MU^6(BG) \otimes_{MU^*} \mathbb{Z}$ onto the *torsion* of the target, are strictly stronger statements about which we say nothing. Nor does it decide the isomorphism half (injectivity questions for cl), or any statement at other primes beyond the classical ones.

Remark 5.4 (The mechanism, and the next cases). The twisted geometry contributes exactly one thing: an odd transfer value, whose oddness traces through the Segre expansion to the binomial $\binom{6}{2} = 15$. The Brauer twist then descends the computation to BG – where representation theory is parity-blocked – while the same twist pays the torsion bill on its own (Remark 4.2). Nothing here is special to six except two computable fingerprints: the odd binomial, and the zero-divisor relation in the mod-two cohomology of the quotient. A first probe of SL_{10}/μ_2 (the construction of Proposition 3.1 and its base-change identity are rank-independent and port verbatim) finds four odd transfer values among the twenty degree-twelve monomials on the $\mathbb{P}^9$ -bundle of the degree-ten versal algebra – the cheapest being $\pi_*(c_4(V_1)c_8(V_1)) = -4725c_3$ – with greatest common divisor 25 over the transfer values. At rank ten the Chern hits are $c_3(\Lambda^2 \mathrm{std}) = 6c_3$ and $c_3(\mathrm{Sym}^2 \mathrm{std}) = 14c_3$ (the general coefficients being $n-4$ and $n+4$), and $\mathrm{gcd}(25, 6, 14) = 1$: for instance $\pi_*(c_4c_8(V_1)) + 783c_3(\Lambda^2 \mathrm{std}) + 2c_3(\mathrm{Sym}^2 \mathrm{std})$ has split-side restriction $-4725 + 4698 + 28 = +1$, an exact-unit candidate. Whether the rank-ten question has the same critical status (the analogues of the parity theorem and of the ku -index computation must be re-proven, not assumed), and whether the mod-two cohomology of $B\mathrm{PSp}_{10}$ cooperates, are open ports of the method, to be treated elsewhere.

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